

International Islamic University Chittagong  
Department of Computer Science and Engineering  
**Course Outline (Outcome-Based Curriculum)**

**Part A**

1. **Course Code :** STAT-2311 ; **ISCED Code :** 0542
2. **Course Title:** Probability and Statistics
3. **Course Type (GE/ Core Course/ Electives/ . . .):** Core Course
4. **Semester:** Third
5. **Academic Session:** Autumn-26
6. **Course Teacher/Instructor:** Dr. Mohammad Manjur Alam (DMMA)
7. **Prerequisite:** STAT-1201(Statistics)
8. **Credit Value:** 2
9. **Contact Hours: 2 Lecture Hours Per Week**
10. **Total Marks:** 100
11. **Rationale of the Course:**

This course is to provide an understanding for the engineering student on statistical concepts to include measurements of location and dispersion, probability, probability distributions, hypothesis testing, regression, and correlation analysis.

**12. Course Objectives:**

- i. Demonstrate understanding of descriptive statistics by practical application of quantitative reasoning and to the solution of engineering problems with data visualization.
- ii. Be able to compute and interpret the results of correlation and regression.
- iii. Probability and its distributions to various engineering problems.
- iv. Perform hypothesis testing using statistical methods and estimation.

**13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)**

| Course Learning Outcomes (CLO) (3-5)                                                                                               | PLO1 | PLO2 | PLO4 |
|------------------------------------------------------------------------------------------------------------------------------------|------|------|------|
| <b>CLO1 (Cognitive – Understand/Apply):</b> Knowledge of descriptive statistics for practical application of quantitative research | ✓    |      |      |
| <b>CLO2 (Cognitive – Apply/Analyze):</b> Analyze various statistical problems used in engineering                                  |      | ✓    |      |
| <b>CLO3 (Cognitive – Apply):</b> Apply probability theory and its distributions to engineering problems                            | ✓    |      |      |
| <b>CLO4 (Cognitive – Create/Evaluate):</b> Create hypothesis for data analysis                                                     |      |      | ✓    |

**Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy**

| CLOs | Teaching-Learning Strategy                                                                           | Assessment Strategy                        |
|------|------------------------------------------------------------------------------------------------------|--------------------------------------------|
| CLO1 | Lectures, tutorial classes, problem-solving sessions, data visualization exercises                   | Class Test 1, Quiz, Assignment-1, Mid Exam |
| CLO2 | Lectures, numerical problem-solving, case studies on engineering data, computer-based practice       | Assignment-2, Final Exam                   |
| CLO3 | Lectures, guided practice on probability problems, real-life engineering examples, group discussion. | Presentation, Final Exam                   |
| CLO4 | Lectures, step-by-step hypothesis testing practice, problem-based learning.                          | Class Test 2, Final Exam                   |

**Part B****14. Course plan specifying content, CLOs, co-curricular activities (if any), Teaching-Learning and Assessment Strategy mapped with CLOs:**

| Week | Topic                                                                                                                                                                                                                               | Teaching-Learning Strategy                                 | Assessment Strategy       | CLOs |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|---------------------------|------|
| 1    | Introduction and overview of statistics, Origin and history of statistics, necessity and importance of statistics in engineering and others field. Population and Sample, Variable and Constants, Different types of variables.     | Lectures, interactive discussion                           | Mid Exam                  | CLO1 |
| 2    | Statistical data, Data Collection and Presentation. Construction of Frequency distribution, Graphical presentation of Frequency distribution.                                                                                       | Lectures, data visualization exercises                     | Quiz, Mid Exam            | CLO1 |
| 3    | Arithmetic Mean, Geometric Mean, Harmonic Mean, Median, Mode. Weighted Mean. Quartile, Deciles, Percentile. Theorems & Problems of measures of central tendency.                                                                    | Lectures, problem-solving sessions                         | Assignment-1, Mid Exam    | CLO1 |
| 4    | Measures of dispersion. Range, Quartile Deviation, Mean Deviation, Variance and Standard Deviation.                                                                                                                                 | Lectures, problem-solving sessions                         | Mid Exam                  | CLO1 |
| 5    | Moments, Skewness and Kurtosis. Problems of Moments, Skewness and Kurtosis.                                                                                                                                                         | Lectures, flipped classroom, video demonstration           | Class Test-1 Mid Exam     | CLO1 |
| 6    | Review of Mid Term syllabus                                                                                                                                                                                                         | Revision, problem-solving practice                         | Mid Exam                  | CLO1 |
| 7    | Concept of Bi-variable data, construction of bi-variate frequency table, Scatter Diagram, Simple Correlation and its measure. Properties of simple correlation coefficient, Spearman's Rank correlation coefficient related theorem | Lectures, interactive discussion                           | Final Exam                | CLO2 |
| 8    | Linear and non-linear regression, properties of regression. Least-square method of curve fittings, related theorems                                                                                                                 | Lectures, data visualization exercises                     | Final Exam                | CLO2 |
| 9    | Problems of Correlation and regression.                                                                                                                                                                                             | Numerical practice, tutorial sessions                      | Final Exam Assignment-2   | CLO2 |
| 10   | Elementary Concepts, Approaches of defining probability, Laws of Probability – Additive and Multiplicative Law, Conditional probability                                                                                             | Lectures, examples, group discussion                       | Final Exam                | CLO3 |
| 11   | Classical, Empirical, Subjective and Axiomatic Probability and independent event. Bay's theorem and related problems.                                                                                                               | Lectures, real-life engineering examples, group discussion | Presentation, Final Exam, | CLO3 |
| 12   | Random Variables, Mathematical Expectation, Variance of a random variable, Theorems & Problems                                                                                                                                      | Lectures, numerical practice                               | Final Exam                | CLO3 |

|    |                                                                                                                                                                                                                       |                                         |                            |                  |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------------------|------------------|
| 13 | Binomial distribution, Poisson distribution and Normal distribution – Their properties, uses, Theorems & Problems.                                                                                                    | Problem-solving, real-life applications | Final Exam                 | CLO3             |
| 14 | Basic ideas of statistical hypothesis, Test regarding single mean, test of equality of two mean, test of equality of several means (Analysis of variance), test regarding proportion. General concepts of Estimation. | Lectures, step-by-step problem solving  | Class Test-2<br>Final Exam | CLO4             |
| 15 | Review of Final Exam syllabus and improvement plan for future outbreaks                                                                                                                                               | Revision, problem-solving practice      | Final Exam                 | CLO2, CLO3, CLO4 |

## Part C

### 15. Assessment and Evaluation

#### 1) Assessment Strategy

| CLOs | Assessment MAP                                     |
|------|----------------------------------------------------|
| CLO1 | Quiz + Class Test-1 + Assignment-1 + Mid-Term Exam |
| CLO2 | Assignment-2 + Final Exam                          |
| CLO3 | Presentation + Final Exam                          |
| CLO4 | Class Test-2 + Final Exam                          |

#### 2) Marks distribution:

##### a) Continuous Assessment (CIE):

| Assessment Tools                             | Weightage (50%) |
|----------------------------------------------|-----------------|
| Class Tests, Assignments, Quiz, Presentation | 10              |
| Mid Term Exam                                | 30              |
| Class Attendance                             | 10              |

##### b) Summative (SEE):

| Assessment Tools   | Weightage (50%) |
|--------------------|-----------------|
| Final Examinations | 50              |

#### 3) Make-up Procedures

The course teacher will schedule the make-up class in consultation with students, ensuring no conflict with other courses, and the timetable will be communicated in advance with the approval of the departmental chairman.

## Part D

### 15. Learning Materials

#### 1) Recommended Readings:

| # | Name of Authors                              | Title of Book                                     | Edition       | Publisher's Name          | Year | ISBN                    |
|---|----------------------------------------------|---------------------------------------------------|---------------|---------------------------|------|-------------------------|
| 1 | Montgomery, Douglas C. and Runger, George C. | Applied Statistics and Probability for Engineers, | Fifth Edition | John Wiley and Sons, Inc. | 2011 | ISBN: 978-0-470-05304-1 |

|   |                                                             |                                                                  |         |               |      |                                   |
|---|-------------------------------------------------------------|------------------------------------------------------------------|---------|---------------|------|-----------------------------------|
| 2 | Walpole, R.E.,<br>Myers, R.H.,<br>Myers, S.L. &<br>Ye, K.Y. | Probability and<br>Statistics for<br>Engineers and<br>Scientist. | 9th ed. | Prentice Hall | 2012 | ISBN-13:<br>978-0-321-<br>74823-9 |
|---|-------------------------------------------------------------|------------------------------------------------------------------|---------|---------------|------|-----------------------------------|

## 2) Supplementary Readings

| #  | Name of Authors         | Title of Book                                                            | Edition       | Publisher's Name           | Year | ISBN                       |
|----|-------------------------|--------------------------------------------------------------------------|---------------|----------------------------|------|----------------------------|
| 1. | Alberto Leon-Garcia     | Probability, Statistics, and Random Processes for Electrical Engineering | 1st Edition   | Cambridge University Press | 2014 | ISBN-13:<br>978-1107439887 |
| 2  | Prof. Dr. M. G. Mostafa | Methods of Statistics                                                    | Reprint, 2016 | Brothers Publications      | 2016 | -                          |

## 3) Others (as applicable for the discipline academic program)

- Lecture notes and handouts provided by the course teacher
- Online resources (e.g., NPTEL, MIT Open Courseware, Khan Academy)  
<https://www.khanacademy.org/math/statistics-probability>
- Statistical software/tools (e.g., MS Excel, SPSS)

**Name of the course teacher: Dr. Mohammad Manjur Alam**

Signature of the Course Teacher

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Date: 19. 8. 2026